Buffon's Needle Experiment

A Statistical Game for Estimating π

Seminar Course

May 14, 2026

What Is Buffon's Needle Experiment?

Definition

Buffon's Needle is a famous probability experiment used to estimate the value of π using random needle drops.

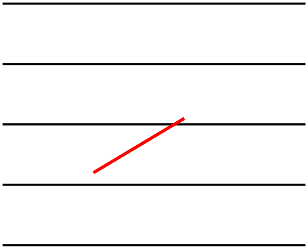
- Monte Carlo Simulation
- Geometric Probability
- Statistical Estimation
- Law of Large Numbers

Historical Note

Introduced by **Georges-Louis Leclerc, Comte de Buffon** in the 1700s.

Materials Needed

- Toothpicks
- Paper with parallel lines
- Ruler
- Calculator
- Pen and paper



Experimental Setup

Simplest Setup

Choose:

$$L = D$$

Where:

- $L =$ Needle length
- $D =$ Distance between lines

Example:

$$L = 4 \text{ cm}, \quad D = 4 \text{ cm}$$

Experimental Setup

Simplest Setup

Choose:

$$L = D$$

Where:

- $L =$ Needle length
- $D =$ Distance between lines

Example:

$$L = 4 \text{ cm}, \quad D = 4 \text{ cm}$$

This simplifies the probability formula significantly.

How To Play

1. Draw equally spaced parallel lines

How To Play

1. Draw equally spaced parallel lines
2. Randomly drop toothpicks

How To Play

1. Draw equally spaced parallel lines
2. Randomly drop toothpicks
3. Record whether each needle crosses a line

How To Play

1. Draw equally spaced parallel lines
2. Randomly drop toothpicks
3. Record whether each needle crosses a line
4. Count crossings

Trial	Crosses a Line?
1	Yes
2	No
3	Yes
4	No

The Mathematics

When:

$$L = D$$

The crossing probability is:

$$P = \frac{2}{\pi}$$

The Mathematics

When:

$$L = D$$

The crossing probability is:

$$P = \frac{2}{\pi}$$

Estimate probability using experimental data:

$$P \approx \frac{C}{N}$$

Where:

- N = Total drops
- C = Crossings

The Mathematics

When:

$$L = D$$

The crossing probability is:

$$P = \frac{2}{\pi}$$

Estimate probability using experimental data:

$$P \approx \frac{C}{N}$$

Where:

- $N =$ Total drops
- $C =$ Crossings

Therefore:

$$\pi \approx \frac{2N}{C}$$

Worked Example

Suppose:

$$N = 100$$

$$C = 63$$

Then:

$$\pi \approx \frac{2(100)}{63}$$

Worked Example

Suppose:

$$N = 100$$

$$C = 63$$

Then:

$$\pi \approx \frac{2(100)}{63}$$

$$\pi \approx 3.17$$

Observation

Very close to the true value:

$$\pi = 3.14159...$$

Why Is This Important?

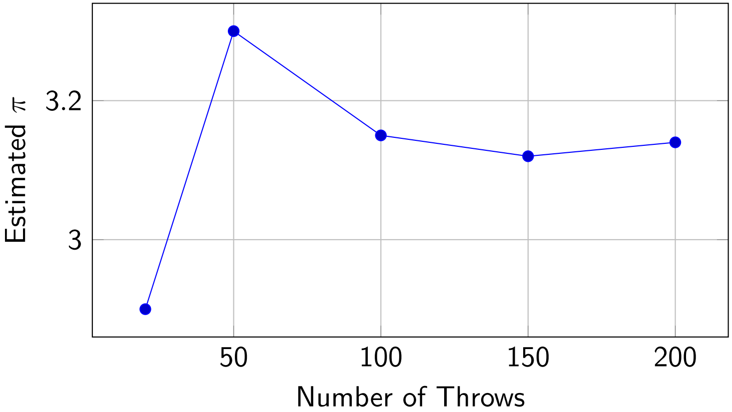
Students Learn

- Randomness
- Simulation
- Estimation
- Probability
- Convergence

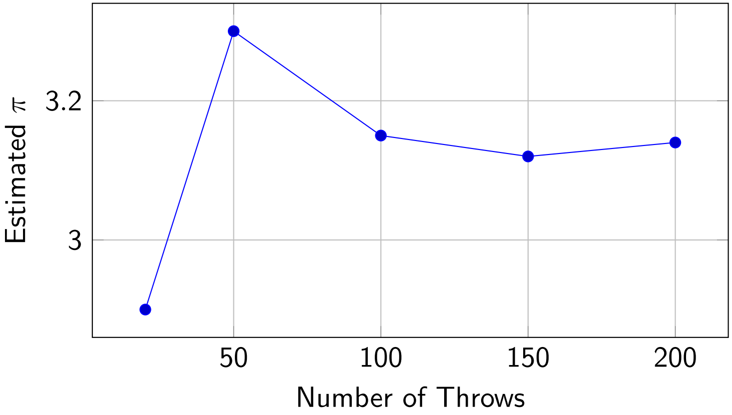
Real Applications

- AI simulations
- Physics modeling
- Finance
- Bayesian statistics
- Machine learning

Law of Large Numbers



Law of Large Numbers



Key Idea

More trials produce estimates closer to the true value.

Classroom Competition Activity

Interactive Group Challenge

Each group performs:

- 50 needle drops
- Estimates π
- Reports results

Classroom Competition Activity

Interactive Group Challenge

Each group performs:

- 50 needle drops
- Estimates π
- Reports results

Winning Categories:

1. Closest estimate to π
2. Most consistent results
3. Best explanation

Statistical Concepts Covered

Concept	How It Appears
Probability	Crossing probability
Simulation	Random needle drops
Estimation	Estimating π
Monte Carlo	Entire experiment
Variability	Group differences
Convergence	Accuracy improves

Fun Discussion Questions

1. Why does randomness estimate π?

Fun Discussion Questions

1. Why does randomness estimate π?
2. Why do larger samples improve accuracy?

Fun Discussion Questions

1. Why does randomness estimate π?
2. Why do larger samples improve accuracy?
3. What causes differences between groups?

Conclusion

Main Takeaway

Buffon's Needle Experiment transforms a simple game into a powerful demonstration of:

- Probability
- Statistical estimation
- Simulation methods
- Monte Carlo techniques
- Real-world statistical thinking

Questions?